

put voltage (after setting accuracy is applied).

Readback resolution. It is the smallest change in internally-measured output voltage or current that is discernible by the power supply. It is usually expressed as an absolute value, but can also be given as a percentage of full scale.

Stability. A power supply's performance inevitably changes due to aging. Maintaining long-term stability demands regular verification and calibration.

Temperature stability. Power supply accuracy is usually specified over a temperature range, often between 20°C and 30°C.

Load regulation. It is a measure of the ability of the output voltage or output current to remain constant during changes in the load. This familiar format is easy to understand and verifiable through testing. It is given as $\pm$ (% of setting + offset).

Line regulation. It is a measure of the ability of the power supply to maintain its output voltage or output current while its AC line input voltage and frequency vary over the full allowable range. This offers a worst-case picture, which is given as $\pm$ (% of setting + offset).

Direct-current power supplies do not actually produce perfect DC outputs. Output AC noise and transient response change both load and settings. Some of the AC characteristics are described below.

Ripple and noise. Spurious AC components on the output of a DC supply are also often referred to as periodic and random deviation (PARD). The term ripple refers to periodic AC on the output. When viewed in frequency domain, ripple shows up as spurious responses. Unlike ripple, noise is random. Noise covers a broad spectrum, and when viewed in frequency domain, manifests itself as an increase in the baseline.

Transient response. It is tested by applying significant step changes

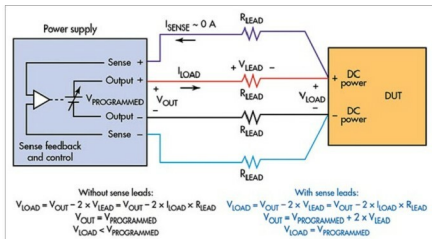


Fig. 2: Remote sense capability

to load impedance and power supply settings, and measuring the time to settle to a stable DC value.

Required output accuracy level

It is important to review the power supply's output accuracy and readback specifications in case of accurate control of voltage at the load for research experimentation or device characterisation.

Programmable power supplies are equipped with remote sense capability. Remote sensing is required in applications where load is located at some distance, typically > 3m (10-feet) from the power supply output terminals. It solves the problem of voltage drop in the leads by extending the power-supply feedback loop to the input of the load.

The difference in voltage is based on the amount of current and the load lead size and length. It uses a 4-wire connection (Fig. 2) to make sure the voltage you set on the supply is the voltage you get at the device under test (DUT) despite voltage drop in cables that carry current between the power supply and DUT.

Acceptable maximum level of noise in the supply's output

For powering a very low-voltage circuit, or a circuit that uses or measures very low currents, such as a transducer detector that must pick

up millivolt or microampere signals, noise from external sources may cause problems.

The power supply itself is a noise source. This noise breaks down into two components: normal mode and common mode. Normal-mode noise, which is generated across the power supply's output terminals, exudes from the power supply's internal circuitry.

Common-mode noise is Earth-referenced noise originating from the power line and stray capacitance across the main transformer.

Two types of bench DC power supplies are commonly used today, namely, linear and switch-mode.

Linear supplies are simple and heavier because the 50Hz or 60Hz transformer and associated filters are physically larger. The linear topology generates minimum noise on the power supply's output.

Switch-mode power supplies (SMPS) are significantly smaller, lighter and more efficient than linear power supplies, so these have replaced linear supplies for higher power requirements.

On the negative side, high switching frequency of SMPS generates five to ten times more noise than a linear supply. Whenever it commands minimising noise, choose a linear supply (if one is available) based on power requirements.